

PreCalculus-Inverse Trig Domain and Range(Learning Target I)

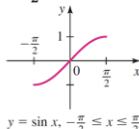
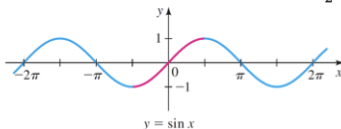
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$\sin^{-1}(x)$ and $\cos^{-1}(x)$

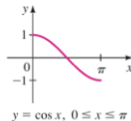
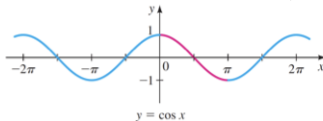
DEFINITION: The **inverse sine function**, denoted by $\sin^{-1} x$ (or $\arcsin x$), is defined to be the inverse of the restricted sine function

$$\sin x, \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$



DEFINITION: The **inverse cosine function**, denoted by $\cos^{-1} x$ (or $\arccos x$), is defined to be the inverse of the restricted cosine function

$$\cos x, \quad 0 \leq x \leq \pi$$

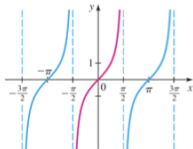


- ▶ For $\sin(x)$, restricting $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ makes $\sin(x)$ to be 1-to-1, so $\sin(x)$ can have its inverse, and $\sin^{-1}x$ has *domain* $[-1, 1]$, and *range* $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- ▶ For $\cos(x)$, restricting $x \in [0, \pi]$ makes $\cos(x)$ to be 1-to-1 so $\cos(x)$ can have its inverse, and $\cos^{-1}x$ has *domain* $[-1, 1]$, and *range* $[0, \pi]$

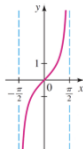
$\tan^{-1}(x)$ and $\cot^{-1}(x)$

DEFINITION: The **inverse tangent function**, denoted by $\tan^{-1}x$ (or $\arctan x$), is defined to be the inverse of the restricted tangent function

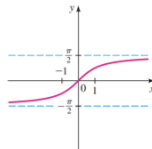
$$\tan x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$



$y = \tan x$



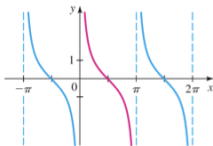
$y = \tan x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$



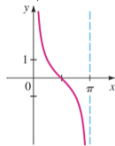
$y = \tan^{-1}x$

DEFINITION: The **inverse cotangent function**, denoted by $\cot^{-1}x$ (or $\operatorname{arccot} x$), is defined to be the inverse of the restricted cotangent function

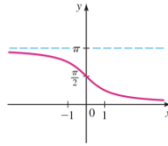
$$\cot x, \quad 0 < x < \pi$$



$y = \cot x$



$y = \cot x, \quad 0 < x < \pi$



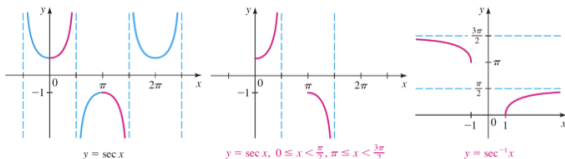
$y = \cot^{-1}x$

- ▶ For $\tan(x)$, restricting $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ makes $\tan(x)$ to be 1-to-1, so $\tan(x)$ can have its inverse, and $\tan^{-1}x$ has domain $= (-\infty, \infty)$, and range $= (-\frac{\pi}{2}, \frac{\pi}{2})$
- ▶ For $\cot(x)$, restricting $x \in (0, \pi)$ makes $\cot(x)$ to be 1-to-1, so $\cot(x)$ can have its inverse, and $\cot^{-1}x$ has domain $= (-\infty, \infty)$, and range $= (0, \pi)$

$\sec^{-1}(x)$ and $\csc^{-1}(x)$

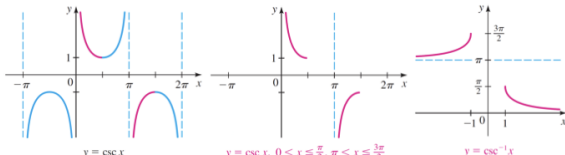
DEFINITION: The **inverse secant function**, denoted by $\sec^{-1} x$ (or $\operatorname{arcsec} x$), is defined to be the inverse of the restricted secant function

$\sec x$, $x \in [0, \pi/2) \cup [\pi, 3\pi/2)$ [or $x \in [0, \pi/2) \cup (\pi/2, \pi]$ in some other textbooks]



DEFINITION: The **inverse cosecant function**, denoted by $\csc^{-1} x$ (or $\operatorname{arccsc} x$), is defined to be the inverse of the restricted cosecant function

$\csc x$, $x \in (0, \pi/2] \cup (\pi, 3\pi/2]$ [or $x \in [-\pi/2, 0) \cup (0, \pi/2]$ in some other textbooks]



- ▶ For $\sec(x)$, restricting $x \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$ makes $\sec(x)$ to be 1-to-1, so $\sec(x)$ can have its inverse, and $\sec^{-1} x$ has *domain* $= (-\infty, -1] \cup [1, \infty)$, and *range* $= [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$
- ▶ For $\csc(x)$, restricting $x \in (0, \frac{\pi}{2}] \cup (\pi, \frac{3\pi}{2}]$ makes $\csc(x)$ to be 1-to-1 so $\csc(x)$ can have its inverse, and $\csc^{-1} x$ has *domain* $= (-\infty, -1] \cup [1, \infty)$, and *range* $= (0, \frac{\pi}{2}] \cup (\pi, \frac{3\pi}{2}]$

Summary

Inverse Function	Domain	Range	Illustration of Range on the Unit Circle
$\sin^{-1}(x)$ or $\arcsin(x)$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	
$\cos^{-1}(x)$ or $\arccos(x)$	$[-1, 1]$	$[0, \pi]$	
$\tan^{-1}(x)$ or $\arctan(x)$	$(-\infty, \infty)$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	
$\csc^{-1}(x)$ or $\operatorname{arccsc}(x)$	$(-\infty, -1] \cup [1, \infty)$	$(0, \frac{\pi}{2}] \cup (\pi, \frac{3\pi}{2}]$	
$\sec^{-1}(x)$ or $\operatorname{arcsec}(x)$	$(-\infty, -1] \cup [1, \infty)$	$[0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$	
$\cot^{-1}(x)$ or $\operatorname{arccot}(x)$	$(-\infty, \infty)$	$(0, \pi)$	